

Lesson 1*The linear regression model: assumptions, OLS, and instruments*

1 The regression model

For a single observation i we observe

$$y_i, \quad x_{1i}, \quad x_{2i}, \quad \dots, \quad x_{ki}, \quad i = 1, \dots, n.$$

The *regression function* is the conditional expectation of y_i given the regressors held at given values:

$$\mathbb{E}(y_i \mid x_{1i}, \dots, x_{ki}) = \mathbb{E}[y_i \mid x_{1i} = x_1^*, x_{2i} = x_2^*, \dots, x_{ki} = x_k^*].$$

Supposing the regression function is linear,

$$\mathbb{E}(y_i \mid x_{1i}, \dots, x_{ki}) = x_{1i}\beta_1 + x_{2i}\beta_2 + \dots + x_{ki}\beta_k = x_i'\beta, \quad x_i = (x_{1i}, \dots, x_{ki})', \quad \beta = (\beta_1, \dots, \beta_k)'$$

Stacking the n observations, with the first subscript the variable and the second the observation,

$$X = \begin{bmatrix} x_{11} & x_{21} & \cdots & x_{k1} \\ \vdots & \vdots & & \vdots \\ x_{1n} & x_{2n} & \cdots & x_{kn} \end{bmatrix}_{n \times k}, \quad \text{row } i \text{ of } X \text{ is } x_i'.$$

2 The linear model in matrix form

With all n observations stacked,

$$\mathbb{E}(y \mid X) = X_{n \times k} \beta, \quad \text{the } X\text{'s are fixed.}$$

Decompose y into its mean plus a deviation from the mean:

$$y = \mathbb{E}(y \mid X) + \underbrace{y - \mathbb{E}(y \mid X)}_u,$$

so that

$$y = \mathbb{E}(y \mid X) + u.$$

Writing the regressors in matrix form,

$$y = X\beta + u, \quad \begin{matrix} u \\ n \times 1 \end{matrix} \sim N\left(\begin{matrix} 0 \\ n \times 1 \end{matrix}, \begin{matrix} \sigma^2 I_n \\ n \times n \end{matrix}\right).$$

3 Covariance structure of the errors

$$\text{Var}(u) = \mathbb{E}[(u - \mathbb{E}(u))(u - \mathbb{E}(u))'] = \mathbb{E}(uu') = \sigma^2 I_n.$$

Written out coordinate by coordinate, with u the $n \times 1$ vector:

$$\mathbb{E}(uu') = \mathbb{E} \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \begin{bmatrix} u_1 & \cdots & u_n \end{bmatrix} = \mathbb{E} \begin{bmatrix} u_1^2 & \cdots & u_1 u_n \\ \vdots & \ddots & \vdots \\ u_n u_1 & \cdots & u_n^2 \end{bmatrix}.$$

In the two-dimensional case the off-diagonal terms come in pairs of opposite sign: $u_1 u_2$ is positive in quadrants 1 and 3 and negative in 2 and 4, so with a symmetric (spherical) distribution they cancel in expectation and the covariance matrix is diagonal.

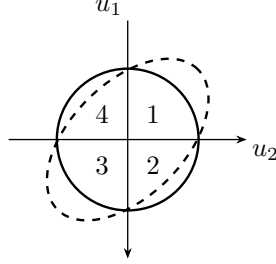


Figure 1: Signs of the cross-product $u_1 u_2$: positive in quadrants 1 and 3, negative in quadrants 2 and 4. Solid: spherical (circular) contour; dashed: a tilted, non-spherical contour, as in the original sketch.

Sphericity assumption

The expectation of this matrix is assumed to be $\sigma^2 I_n$:

$$\mathbb{E}(uu') = \begin{bmatrix} \sigma^2 & 0 & 0 & \cdots & 0 \\ 0 & \sigma^2 & 0 & \cdots & 0 \\ 0 & 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \sigma^2 \end{bmatrix}.$$

That is, the errors have a common variance and are uncorrelated with one another.

4 The rank of the regressor matrix

$$\underset{n \times k}{X} \longrightarrow \text{rank}(X) = k,$$

i.e. X has full column rank. Then

$$\underset{k \times n}{X'} \underset{n \times k}{X} \longrightarrow k \times k \quad \text{with} \quad \text{rank } k,$$

so $X'X$ is a full-rank $k \times k$ matrix and can be inverted.

5 Norms: spherical versus non-spherical covariance

For the model $y = X\beta + u$, $u \sim N(0, \sigma^2 I)$, we have $\mathbb{E}(u) = 0$, so

$$\|u - \mathbb{E}(u)\| = \|u\| = (u'u)^{1/2} = \left(\sum_{i=1}^n u_i^2\right)^{1/2},$$

the ordinary Euclidean norm: every direction is treated alike, which is what $\Sigma = \sigma^2 I_n$ means.

Now suppose instead

$$w \sim N(0, \Sigma_n), \quad \Sigma_n \neq \sigma^2 I_n.$$

Distances are then measured in the metric of Σ^{-1} , so the squared norm becomes

$$\|w\|^2 = w'w \longrightarrow \|w\|_{\Sigma^{-1}}^2 = w'\Sigma^{-1}w.$$

Directions in which w varies a lot are down-weighted, directions in which it varies little are up-weighted. When $\Sigma = \sigma^2 I_n$ this is just $w'w/\sigma^2$, the Euclidean case rescaled. This weighted norm is what is applied below to $X'u$, whose covariance $\sigma^2 X'X$ is not spherical.

6 The least-squares estimator

Minimising the squared length of the residual vector $y - Xb$,

$$\min_b \|y - Xb\|^2 = \min_b (y - Xb)'(y - Xb).$$

Expanding the quadratic form (the two middle terms are the same scalar, since $y'Xb = (b'X'y)'$),

$$\begin{aligned} &= (y' - b'X')(y - Xb) \\ &= y'y - b'X'y - y'Xb + b'X'Xb \\ &= y'y - 2b'X'y + b'X'Xb. \end{aligned}$$

The first-order condition is

$$\frac{\partial}{\partial b} (y'y - 2b'X'y + b'X'Xb) = -2X'y + 2X'Xb = 0,$$

hence the *normal equations* $X'Xb = X'y$ and

$$b = (X'X)^{-1}X'y,$$

which exists because $\text{rank}(X) = k$ makes $X'X$ invertible. The second-order condition is

$$\frac{\partial^2}{\partial b \partial b'} (y'y - 2b'X'y + b'X'Xb) = 2X'X,$$

which is positive definite ($a'X'Xa = \|Xa\|^2 > 0$ for $a \neq 0$, again by full column rank), so b is the minimum.

7 The distribution of $X'u$

Starting from the estimator

$$b = (X'X)^{-1}X'y,$$

y is random. Take the model again,

$$y = X\beta + u, \quad u \sim (0, \sigma^2 I_n),$$

and the exogeneity assumption

$$\mathbb{E}(X'u) = 0.$$

Because the mean is zero, the norm of $X'u$ is its own deviation, and the non-spherical norm of the previous section gives the quadratic form

$$\|X'u - \mathbb{E}(X'u)\| = \|X'u\| \implies (X'u)'[\text{Var}(X'u)]^{-1}X'u.$$

The middle piece is

$$\begin{aligned} \text{Var}(X'u) &= \mathbb{E}[(X'u)(X'u)'] \\ &= \mathbb{E}(X'uu'X) = X'\mathbb{E}(uu')X = \sigma^2 X'X, \end{aligned}$$

taking X outside the expectation because it is fixed. Substituting $\text{Var}(X'u) = \sigma^2 X'X$ into the quadratic form,

$$(X'u)'[\text{Var}(X'u)]^{-1}X'u = (X'u)'(\sigma^2 X'X)^{-1}X'u = \frac{1}{\sigma^2} u'X(X'X)^{-1}X'u.$$

If moreover $u \sim N(0, \sigma^2 I_n)$, then $X'u \sim N(0, \sigma^2 X'X)$, a k -vector, and this quadratic form is distributed χ_k^2 . Replacing the unknown u by $y - Xb$ and minimising over b is exactly the problem of the next section.

8 The same estimator, through the projection matrix

Define the projection matrix

$$P_X = X(X'X)^{-1}X',$$

which is the matrix in the middle of the quadratic form above, and minimise (dropping the constant σ^{-2})

$$\begin{aligned} &\min_b (y - Xb)'X(X'X)^{-1}X'(y - Xb) \\ &= \min_b (y - Xb)'P_X(y - Xb) \\ &= (y' - b'X')P_X(y - Xb) \\ &= y'P_Xy - b'X'P_Xy - y'P_XXb + b'X'P_XXb \\ &= y'P_Xy - 2b'X'P_Xy + b'X'P_XXb. \end{aligned}$$

The first-order condition is

$$\text{FOC: } -2X'P_Xy + 2X'P_XXb = 0,$$

so

$$X'P_Xy = X'P_XXb.$$

Since $P_XX = X(X'X)^{-1}X'X = X$ and P_X is symmetric, $X'P_X = (P_XX)' = X'$, and the condition is the normal equations $X'y = X'Xb$ again. Solving for b gives again

$$b = (X'X)^{-1}X'y.$$

9 When X is not exogenous: instruments

With the model

$$y = X\beta + u, \quad u \sim (0, \sigma^2 I),$$

the exogeneity assumption fails:

$$\mathbb{E}(X'u) \neq 0.$$

Then we look for an instrument matrix

$$Z_{n \times p}, \quad p \geq k,$$

satisfying

$$\mathbb{E}(Z'u) = 0 \quad (\text{validity}), \quad \text{rank}(\mathbb{E}(Z'X)) = k \quad (\text{relevance}).$$

Both are needed to identify β : validity gives the p moment equations $\mathbb{E}(Z'(y - X\beta)) = 0$, and relevance makes them pin down a unique β . The case $p = k$ is *exactly identified*, $p > k$ *over-identified*.

Transcription notes

- Page 1 of the original carries only the title “Econometric I”.
- The original mixes T and $'$ for transposes ($x_i^T \beta$ on page 2, $\cdot'$ elsewhere); the prime is used throughout here.
- The regressor matrix is written with the variable as the first subscript (x_{1i} = variable 1, observation i), so row i of X is x'_i ; that convention is kept.
- **Corrected:** page 3 has $\mathbb{E}(y | x) = x^T \beta$ with x marked $n \times k$; for an $n \times k$ matrix the conditional mean of the n -vector y is $X\beta$ ($x^T \beta$ would not conform), so it is typeset $\mathbb{E}(y | X) = X\beta$.
- **Corrected:** the regressor matrix on page 2 has first row $x_{11}, x_{21}, \dots, x_{k1}$ (variable first) but last row $x_{n1}, \dots, x_{nk}$ (observation first); the last row is typeset $x_{1n}, \dots, x_{kn}$ to match the variable-first convention.
- Page 2 also sketches row i of X as $[x_{1i} \ x_{2i} \ \cdots \ x_{ki}]$; this is the remark “row i of X is x'_i ”.
- **Corrected:** $\text{Var}(u)$ was written as $\mathbb{E}(u - \mathbb{E}(u))(u - \mathbb{E}(u))'$; without the outer bracket the operator $\mathbb{E}$ applies only to the first factor, so the expectation is taken over the whole product.
- **Corrected:** “ u ” was labelled by a brace under $y - \mathbb{E}(y | x)$; it is the residual, and it is kept as the definition of u rather than written as a separate assertion.
- **Corrected:** the norm line reads $(u'u)^{1/2} = \sum_{i=1}^n u_i^2$ in the original; the outer square root was missing from the sum.
- **Corrected:** the minimisation was written $\min \|w\|$, while the right-hand side is the *squared* norm (the sum of squared residuals); it is typeset as $\|y - Xb\|^2$, since w was already used for the non-spherical error vector of the previous section. The original also minimises over β and then differentiates in b ; b is used throughout.
- **Corrected:** the weighted norm was written $\|w\| \rightarrow w'\Sigma^{-1}w$; $w'\Sigma^{-1}w$ is the *squared* weighted norm, and it is typeset so.
- **Corrected:** page 6 writes the non-spherical case as $\Sigma_n \neq I_n$; it is typeset $\Sigma_n \neq \sigma^2 I_n$, since a common variance other than one is still spherical.
- **Added:** the second-order condition for least squares, the step $X'P_X = X'$ that turns the projection-matrix first-order condition back into the normal equations, the χ_k^2 distribution of the quadratic form in $X'u$ under normality, and the relevance condition $\text{rank } \mathbb{E}(Z'X) = k$ for the instruments (the original states only $\mathbb{E}(Z'u) = 0$).

- The projection-matrix derivation now follows the distribution of $X'u$, as in the original (pages 8 and 9): the quadratic form $\sigma^{-2}u'P_Xu$ is what motivates minimising $(y - Xb)'P_X(y - Xb)$.
- **Corrected:** the last line of the first-order condition was written as an equality with the preceding expression ($\dots = y'y - 2b'x'y + b'x'xb = -2x'y + 2x'xb = 0$). The derivative is a separate step, so it is typeset as $\partial/\partial b$ of that expression.
- **Corrected:** in the projection-matrix section one line lost the transpose, “ $(y - xb)P_x(y - xb)$ ”; it is $(y - Xb)'P_X(y - Xb)$. The definition $P_X = X(X'X)^{-1}X'$ is written as an equation (it was a label under the expression).
- **Corrected:** the quadratic form for $x'u$ was left with a bare “=” at the front of the last line; it is written here as the full chain $(X'u)'[\text{Var}(X'u)]^{-1}X'u = (X'u)'(\sigma^2X'X)^{-1}X'u = \sigma^{-2}u'X(X'X)^{-1}X'u$, with the substitution step shown.
- The original writes the regressor matrix in lower case (x, P_x) on pages 3 and 6–9 and in upper case (X) elsewhere; it is X (and P_X) throughout here. Transcript notes that quote the original keep its x .
- The expansion of $\mathbb{E}(uu')$ was drawn with the bottom-left entry as u_nu_1 ; the matrix is symmetric (u_iu_j off the diagonal).
- The quadrant sketch was redrawn, keeping its axes u_1 (vertical), u_2 (horizontal) and the quadrant numbers 1–4 (clockwise from top right). The original has a circle and a second, tilted closed curve over it; the latter is drawn dashed and read as a non-spherical contour (interpretation).
- **Added:** the paragraph on the signs of u_1u_2 by quadrant, and the remarks in the norms section on down-/up-weighting of directions; the original has only the sketch and the formulas.
- In the original the expansion of $\mathbb{E}(uu')$ ends with an arrow to the σ^2 -diagonal matrix labelled “sphericity assumption”; the matrix is given once, under that heading, rather than twice.
- Page 7: the last line reads “ $= 2x'y + 2x'xb = 0$ ” with the sign of $2x'y$ unclear; the derivative is $-2X'y + 2X'Xb$.
- Page numbers of the original, in case you need to trace back: page 2 regression function and X ; page 3 the model and σ^2I_n ; page 4 $\text{Var}(u)$ and the sketch; page 5 sphericity and $\text{rank}(X)$; page 6 norms and non-spherical Σ_n ; page 7 least squares; page 8 the distribution of $x'u$; page 9 the projection-matrix derivation; page 10 instruments.